# SIMATS ENGINEERING
## ASSESSMENT TOOL 1

**COURSE NAME: COMPUTER NETWORKS**

**COURSE CODE: CSA0706**

**COURSE OUTCOME COVERED**

***CO5:***

Design network topologies star, mesh, bus, and hybrid using networking devices, transmission media, and cabling standards (*BL2, BL5)*

*Assessment Tool Weightage: 40%*

**QUESTIONS: 3** **Total Marks:30**

**Annexure A**

## Constraint-Based Problem Solving

***Code of Conduct:***

I **M Nishanth (192511154)** certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

***Signature of the student*: M Nishanth**

## Annexure A

# Constraint-Based Problem Solving

1. A multinational company is opening a new branch office in a remote rural location where high-speed wired internet is unavailable. The office must support cloud-based applications, video conferencing, and secure communication with the headquarters. The company has a limited budget for connectivity but requires uninterrupted business operations. As a network engineer, recommend a suitable internet connection, networking devices, and security mechanisms. Justify your recommendations by considering bandwidth limitations, reliability, cost, and future scalability

2. A university plans to provide campus-wide wireless internet access for more than 3,000 students across classrooms, laboratories, libraries, and hostels. The network must support a large number of simultaneous users while minimizing signal interference and ensuring secure access. Due to budget constraints, only a limited number of wireless access points can be deployed. Analyze the requirements and propose an efficient wireless network design, including access point placement, frequency band selection, and authentication methods. Explain how your solution balances coverage, performance, security, and cost.

3. A financial institution is upgrading its internal computer network to improve cybersecurity while allowing employees to access sensitive customer information and online banking services. The organization must comply with strict security policies without significantly affecting network performance or increasing operational costs. Design a secure network architecture by selecting appropriate network segmentation techniques, firewall deployment, authentication methods, and intrusion detection mechanisms. Justify your design by considering the constraints of security, performance, compliance, and budget.

**SIMATS ENGINEERING**

**ASSESSMENT TOOL 1**

## Annexure A — Constraint-Based Problem Solving

### Question 1 (10 Marks) — Rural Branch Office Connectivity

A multinational company is opening a branch in a remote rural location without high-speed wired internet, needing to support cloud applications, video conferencing, and secure communication with headquarters, on a limited budget, with uninterrupted operations required.

**1. Understanding the Problem and Constraints**

- **Connectivity constraint:** No wired broadband/fibre available in the area.
- **Application constraint:** Must reliably run cloud apps (SaaS/ERP), video conferencing (bandwidth- and latency-sensitive), and secure VPN traffic to HQ.
- **Budget constraint:** Limited capital and operating expenditure — solution must avoid expensive last-mile fibre trenching.
- **Reliability constraint:** Business operations cannot be interrupted, so a single point of failure is unacceptable.
- **Scalability constraint:** The office may add staff/services later, so the design must grow without a full re-architecture.

**2. Problem Decomposition**

The problem is broken into four independently solvable sub-problems:

- Sub-problem A — Last-mile Internet access (how to get bandwidth into the building).
- Sub-problem B — Redundancy/failover (how to avoid downtime).
- Sub-problem C — Local networking devices/topology (how traffic moves inside the office).
- Sub-problem D — Security for cloud and HQ communication.

**3. Proposed Solution**

**Sub-problem A — Internet access:**

Deploy a business-grade 4G/5G LTE fixed-wireless connection as the primary WAN link, supplemented by a VSAT (satellite) or point-to-point microwave link where cellular coverage is weak. LTE/5G routers (e.g., Cradlepoint/MikroTik LTE CPE) avoid the cost of trenching fibre while still delivering tens of Mbps, sufficient for video conferencing and cloud access.

**Sub-problem B — Redundancy:**

Use a dual-WAN router/SD-WAN appliance with automatic failover between the 4G/5G link (primary) and a secondary link — either a second SIM from a different carrier or a low-cost satellite backup. SD-WAN also performs intelligent path selection, sending latency-sensitive video traffic over the better link in real time.

**Sub-problem C — Local topology and devices:**

Inside the branch, deploy a star topology: a Layer 3 switch as the core connected to the SD-WAN/dual-WAN router, with access switches and Wi-Fi access points as the spokes. Star topology is chosen because a single cable fault only isolates one device, it is easy to troubleshoot, and it scales by simply adding another switch port — all important given the limited on-site IT support in a remote branch.

**Sub-problem D — Security:**

Terminate a site-to-site IPsec VPN (or a lightweight SD-WAN overlay with built-in encryption) between the branch router and HQ so all traffic to headquarters is encrypted. Use a Unified Threat Management (UTM) firewall at the branch edge for stateful inspection, intrusion prevention, and content filtering, and enforce WPA3-Enterprise with 802.1X on the Wi-Fi so only authenticated staff devices join the network.

### 4. Justification Against Constraints

| Constraint | How the Design Addresses It |
|---|---|
| Bandwidth | 4G/5G + SD-WAN aggregation gives enough throughput for video calls and cloud apps; SD-WAN can bond both links for extra capacity during peak use. |
| Reliability | Dual-WAN failover removes the single point of failure; the office stays online even if one carrier's signal drops. |
| Cost | Wireless WAN avoids expensive fibre trenching in a rural area; a Layer 3 switch + APs is far cheaper than a full structured-cabling rollout. |

| Constraint | How the Design Addresses It |
| --- | --- |
| Security | Site-to-site VPN + UTM firewall + WPA3-Enterprise protects data in transit and controls network access without needing on-site security staff. |
| Future scalability | Star topology and a Layer 3 core switch allow new switches, APs, or a second SD-WAN link to be added without redesigning the network. |

**5. Conclusion**

A wireless-first, SD-WAN-managed branch design with a redundant dual-WAN link, a star-topology LAN, and VPN/UTM-based security meets the company's need for uninterrupted, secure, cloud-ready connectivity within a limited budget, while remaining straightforward to scale as the branch grows.

**Question 2 (10 Marks) — Campus-Wide Wireless Network for 3,000+ Students**

A university needs campus-wide Wi-Fi across classrooms, labs, libraries, and hostels for 3,000+ students, supporting many simultaneous users, minimizing interference, ensuring secure access, and using only a limited number of access points (APs) due to budget.

**1. Understanding the Problem and Constraints**

- **Scale constraint:** 3,000+ concurrent users across multiple buildings and building types.
- **Interference constraint:** Dense AP deployment can cause co-channel interference if not planned.
- **Budget constraint:** Only a limited number of APs can be purchased/deployed.
- **Security constraint:** Only authorized students/staff should access the network; guest access must be isolated.
- **Performance constraint:** Must support high-density usage (labs, libraries) and lower-density usage (hostel rooms) differently.

**2. Problem Decomposition**

- Sub-problem A — AP placement and capacity planning for high- vs low-density zones.
- Sub-problem B — Frequency band and channel plan to minimize interference.

- Sub-problem C — Authentication and access control (security).
- Sub-problem D — Centralized management for cost-effective scaling.

### 3. Proposed Solution

**Sub-problem A — AP placement:**

Classify zones by user density: classrooms and libraries (high density, many devices per room) get one AP per room or per two adjoining rooms mounted on the ceiling for even coverage; large lecture halls get sectorized/high-capacity APs; hostels (lower simultaneous density per AP, but many walls) use one AP per 2–3 rooms or per floor corridor. This zone-based allocation concentrates the limited AP budget where user density is highest instead of spreading APs evenly everywhere.

**Sub-problem B — Frequency and channel plan:**

Use dual-band 802.11ac/ax APs operating primarily on the 5 GHz band (more channels, less congestion, ideal for dense classrooms/labs) with 2.4 GHz kept available for legacy/IoT devices and areas needing longer range (hostel corridors). Adjacent APs are assigned non-overlapping channels (e.g., 1/6/11 on 2.4 GHz; non-overlapping 20/40 MHz channels on 5 GHz) in a honeycomb pattern, and transmit power is tuned down in dense zones so cells do not overlap excessively — this directly minimizes co-channel interference.

**Sub-problem C — Authentication/security:**

Deploy WPA3-Enterprise with 802.1X, authenticating each student against the university's RADIUS server tied to their existing student ID/LDAP credentials — this gives per-user (not shared-password) access and lets compromised accounts be revoked individually. A separate, rate-limited guest SSID with a captive portal and VLAN isolation is used for visitors, keeping them off the academic network.

**Sub-problem D — Centralized management:**

Use a cloud/controller-based Wi-Fi system (e.g., a wireless LAN controller or cloud-managed APs) so all APs share one SSID configuration, load-balance clients across channels/APs, and can be monitored and re-tuned centrally — this keeps operational cost low even as the AP count is limited.

#### 4. Justification Against Constraints

| Constraint | How the Design Addresses It |
|---|---|
| Coverage & density | Zone-based AP allocation puts more capacity in high-traffic areas (labs, libraries) instead of wasting budget on uniform deployment. |
| Interference | 5 GHz-first design with a non-overlapping channel plan and power tuning keeps adjacent AP cells from clashing. |
| Security | WPA3-Enterprise + 802.1X/RADIUS gives per-user authentication; guest VLAN isolates non-student traffic. |
| Cost | Centralized/cloud AP management reduces the number of APs needed per user through load balancing, and cuts IT staffing cost for configuration. |
| Performance | Band steering pushes capable devices to the less-congested 5 GHz band, improving throughput for the majority of users. |

#### 5. Conclusion

A density-aware AP placement plan combined with a disciplined 5 GHz-first channel plan, WPA3-Enterprise/802.1X authentication, and centralized controller-based management delivers secure, high-performance Wi-Fi to 3,000+ students within a limited AP budget.

### Question 3 (10 Marks) — Secure Network Architecture for a Financial Institution

A financial institution must upgrade its internal network to improve cybersecurity for employees accessing sensitive customer data and online banking, while complying with strict security policies, and without hurting performance or raising costs significantly.

#### 1. Understanding the Problem and Constraints

- **Security constraint:** Sensitive customer data and banking systems must be protected from unauthorized access and breaches.
- **Compliance constraint:** Must satisfy regulatory standards (e.g., PCI-DSS-style controls) for financial data handling.

- **Performance constraint:** Security controls must not introduce unacceptable latency to transaction processing.
- **Budget constraint:** Operational costs must remain controlled.

**2. Problem Decomposition**

- Sub-problem A — Network segmentation (limiting what a breach can reach).
- Sub-problem B — Perimeter and internal firewall deployment.
- Sub-problem C — Authentication and access control.
- Sub-problem D — Intrusion detection/prevention and monitoring.

**3. Proposed Solution**

**Sub-problem A — Segmentation:**

Use VLANs to separate the network into isolated zones: a DMZ for the public-facing online-banking web/application servers, a core banking/database zone, a general employee/corporate zone, and a management zone for network admin traffic. Inter-VLAN traffic is only permitted through the firewall, so a compromise in the employee zone cannot directly reach the core banking database — this is the primary control satisfying compliance requirements for data segregation.

**Sub-problem B — Firewall deployment:**

Deploy a Next-Generation Firewall (NGFW) at the internet edge protecting the DMZ, and a second internal firewall (or Layer 3 switch with strict ACLs) between the corporate zone and the core banking zone — a defense-in-depth, two-tier firewall design. The NGFW performs deep packet inspection and application-aware filtering so encrypted banking traffic is still checked without a large performance penalty, since inspection happens once at the boundary rather than per host.

**Sub-problem C — Authentication:**

Enforce role-based access control (RBAC) so employees only reach the systems their job requires, combined with multi-factor authentication (MFA) for anyone accessing core banking systems or remote VPN access. Privileged/admin access additionally goes through a jump/bastion host so admin sessions are isolated and logged.

**Sub-problem D — Intrusion detection and monitoring:**

Place an Intrusion Detection/Prevention System (IDS/IPS) inline at the DMZ and core-zone boundaries to flag and block known attack patterns in real time, feeding alerts into a SIEM

(Security Information and Event Management) system for centralized log correlation — this satisfies audit/compliance logging requirements while giving early warning of intrusions.

**4. Justification Against Constraints**

| Constraint | How the Design Addresses It |
| --- | --- |
| Security | Segmentation + two-tier firewalls + IDS/IPS give layered (defense-in-depth) protection so no single control failure exposes customer data. |
| Performance | Inspection is done once at zone boundaries (not per-host), and RBAC/MFA add only a small, one-time authentication step, keeping transaction latency low. |
| Compliance | VLAN segregation of the core banking zone, MFA, and SIEM logging map directly to standard financial data-protection and audit requirements. |
| Budget | Reusing VLANs/ACLs on existing Layer 3 switches for internal segmentation avoids buying a firewall for every zone; only the DMZ and core boundary need dedicated NGFW/IDS appliances. |

**5. Conclusion**

A segmented, zone-based architecture with a two-tier NGFW deployment, RBAC with MFA, and IDS/IPS feeding a SIEM gives the financial institution layered security and compliance-ready auditing, while keeping performance impact and cost low by concentrating heavy inspection only at zone boundaries.

## RUBRICS FOR CALCULATION

| Criteria | Weightage (%) | Exemplary (4) | Proficient (3) | Developing (2) | Beginning (1) |
|---|---|---|---|---|---|
| **Understanding of Problem & Constraints** | 20% | Clearly identifies all constraints and fully understands the problem requirements | Identifies most constraints with minor gaps in understanding | Identifies some constraints but misses key aspects | Shows little or no understanding of constraints or problem |
| **Problem Decomposition** | 15% | Breaks problem into logical, well-structured sub-problems effectively | Breaks problem into sub-parts with some logical flow | Attempts decomposition but lacks clarity or structure | No clear decomposition of the problem |
| **Application of Constraints in Solution** | 20% | Applies all constraints correctly and consistently in solution design | Applies most constraints correctly with minor errors | Applies some constraints but with noticeable errors | Fails to apply constraints appropriately |
| **Logical Reasoning & Strategy** | 15% | Uses strong, clear, and efficient reasoning strategies | Uses generally sound reasoning with minor gaps | Reasoning is partially correct but inconsistent | Reasoning is unclear or incorrect |
| **Solution Accuracy & Feasibility** | 15% | Solution is fully correct, feasible, and optimized | Solution is mostly correct and feasible | Solution has errors or is partially feasible | Solution is incorrect or not feasible |
| **Creativity & Innovation** | 5% | Demonstrates highly innovative and unique approach | Shows some creativity in approach | Limited creativity shown | No creativity; relies on basic or copied ideas |
| **Clarity of Presentation** | 10% | Solution is clearly presented with excellent organization and explanation | Presentation is clear with minor issues | Presentation lacks clarity or organization | Poorly presented and difficult to understand |